

MINE E 630: Assignment #4

Due Date: March 12, 2026 (until 11: 59 pm)

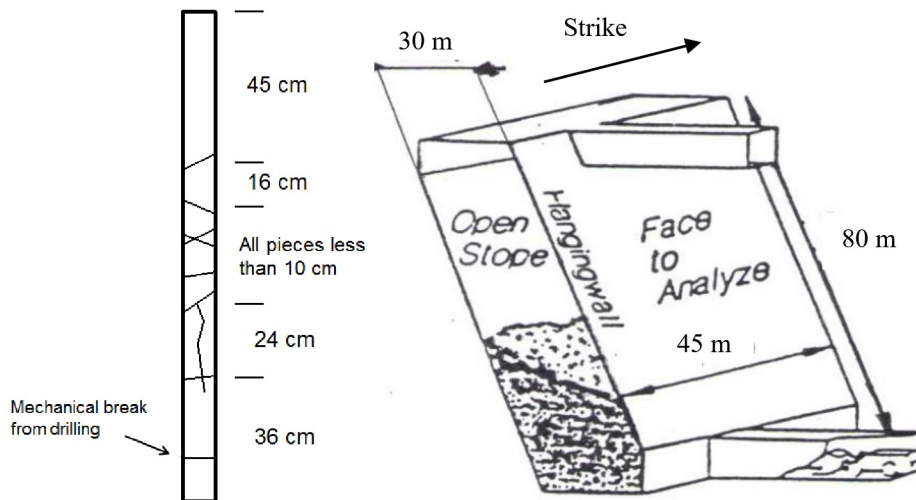
Method of Submission: Please write down your words and submit a combined pdf file to the Canvas assignment before the specified due time.

1. Open Stope design (Please submit your question sheets together with your answer sheets)

An open stope is planned at a depth of 1500 m in an orebody that is 30 m wide and dips at 65°. Because of operational constraints, the preferred stope length is 45 m, and the stope height is 80 m.

The average unconfined compressive strength of the intact rock is 100 MPa. There is one joint set dipping 20° (closely spaced, average spacing is 10 cm) with the same strike direction as the orebody strike.

A typical drill core (the total length of the core is 200 cm) and the stope geometry have been shown below:



Rock mass quality data has been presented in the following table (Question marks in the table are to be answered):

Item	Description	Value	RQD = (45+16+24+36)/200 * 100% = 60.5%
Rock quality	? Fair	?	
Joint sets	One joint set	? 2	
Joint roughness	Smooth undulating	? 2	
Joint alteration	Unaltered with surface staining	? 1	

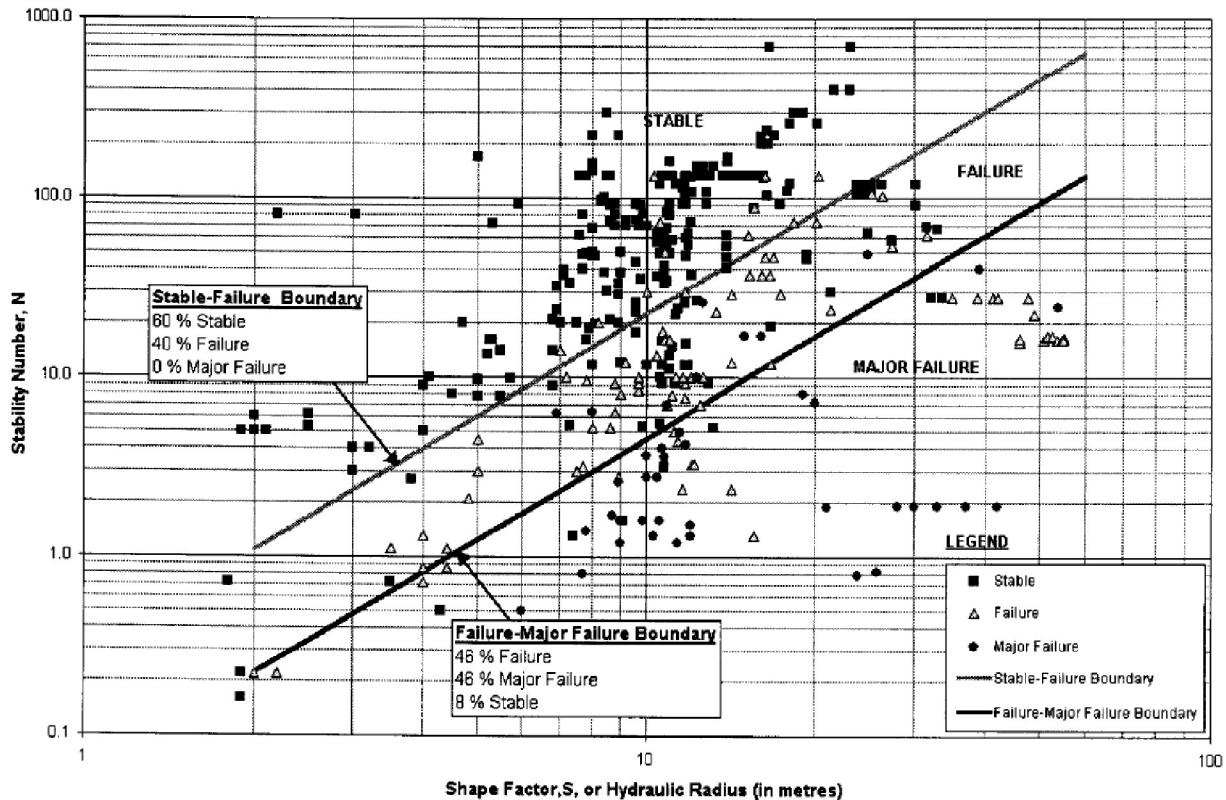
$$\sigma_{H1} = 1.1\sigma_{H2}$$

Along to the strike direction

$$K = \frac{\sigma_{H1}}{\sigma_V} = 1.6 \quad \text{perpendicular to the } \sigma_{H1}$$

The average specific gravity of overburden above the slope is 2.3.

Please estimate the stability of this open slope from the attached Matthews stability graph. Please mark the point and name of each slope surface on this following Matthews Chart with a colour pen.

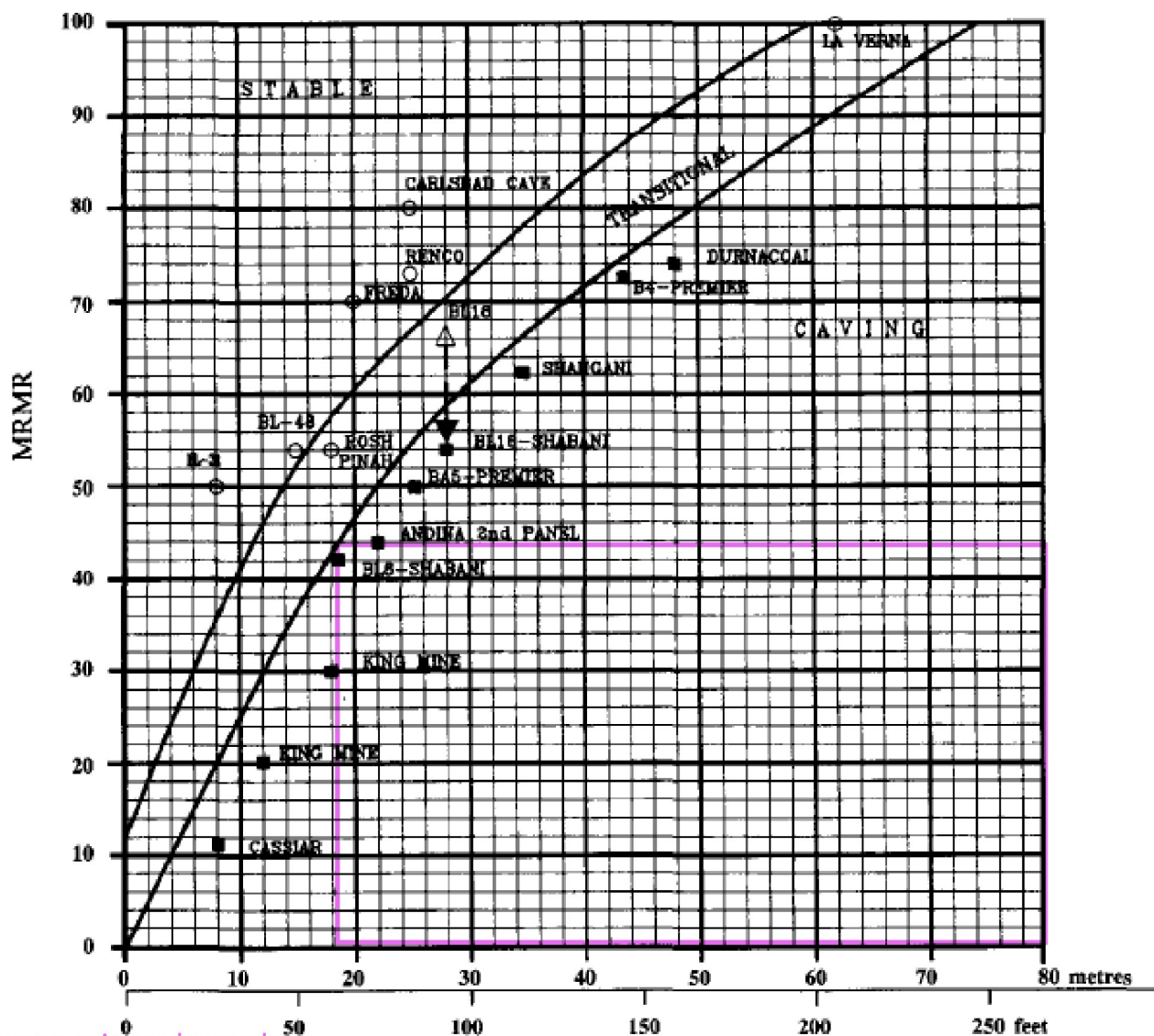


2. Caveability

The Laubscher Stability Chart relates the rock quality and stress conditions for a given deposit, characterized by the Modified Rock Mass Rating (MRMR), to the hydraulic radius of the opening. If **block caving** is considered as the mining method, please estimate the minimum undercutting area (length: width = 1.5) to be required. Please get numbers for all question marks, illustrate your calculations and show your procedure in the attached Laubscher Stability Chart.

Parameter	Description	Rating
UCS	90 MPa	? 9
RQD	80%	? 12
FF/m	3 joint sets, average spacing of 0.5m	? $25 \times 0.7 = 17.5$
Joint condition		? $= 40 \times 0.85 \times 0.9 \times 0.7 = 21.42$
Large scale	Dry, curved	85%
Small scale	Dry, smooth stepped	90%

Joint wall alteration	None	
Joint filling	Soft sheared coarse material	70%
IRMR		? $9+12+17.5+21.42=59.92$
Adjustments		
Weathering	None	100%
Joint orientation	3 joints, 3 inclined	70%
Mining induced stresses		
Blast effects	None	100%
MRMR		? $59.92*1*0.7*1=41.94$



To cave mineral needs
 $HR > 18$.

HYDRAULIC RADIUS = Area/perimeter

So, set width = x , length = $1.5x$ $1.5x/5 > 18 \rightarrow x > 60$, set
 $x=60$ thus the length = 90, so the minimum threshold area is
 $5400m^2$